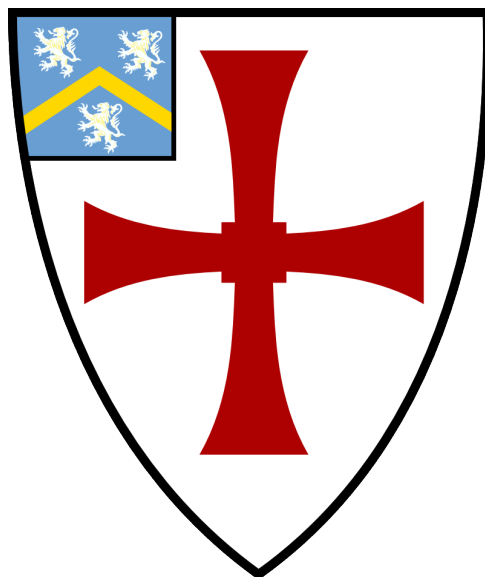


Deep Learning Insights into the Structural Complexity of HIV-1 Capsid Proteins

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Exploring Conformational Space of HIV-1 Capsid Protein with Molearn: A Deep Learning Approach to Structural Bioinformatics

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Abstract

The dynamic nature of proteins plays a pivotal role in nearly every biological process, from molecular recognition to protection of viral genetic material. In this thesis, we leveraged the deep learning power of Molearn to sample conformational space of a viral capsid protein. The research was began by testing the Python repository developed by the Degiacomi Group with a well-studied MurD protein, structurally altered to resemble a multimer. This experiment has proven Molearn can still operate on molecules with coordinate interruptions i.e multimers. Subsequently, Molearn was trained on the largest protein ensemble to date, the hexameric HIV-1 capsid protein. The network performance was evaluated against multiple factors, demonstrating how it scales linearly with the compression factor of the input data. The investigation was finished with a study on latent dimension influence on performance of the autoencoder, on both MurD and the capsid protein, showcasing that latent dimension 7 results in best reconstruction of the latter, while the former is best captured by latent dimensions 3 and 6.

Supervisor: Matteo T. Degiacomi

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1 Introduction

1.1 Proteins

‘Proteins hold the key to the whole subject of the molecular basis of biological reactions’ — *Linus Pauling, Signs of Life*[1]

Proteins are the building blocks of life. They are fundamental macro-molecules which perform undoubtedly crucial tasks in structure and regulation of living organisms[3]. Proteins are the workhorse parts of the cell and are as diverse as the functions they serve, from transport and storage of nutrients, to our immune system’s secret weapon. These polymers show amazingly complex behavior, despite small size and composition of just 20 different amino acid residues linked together via a peptide bond. The exact arrangement of those building blocks solely determines 3 dimensional structure of the protein, thus it’s function in the body. The attainment of protein’s final three-dimensional structure is a

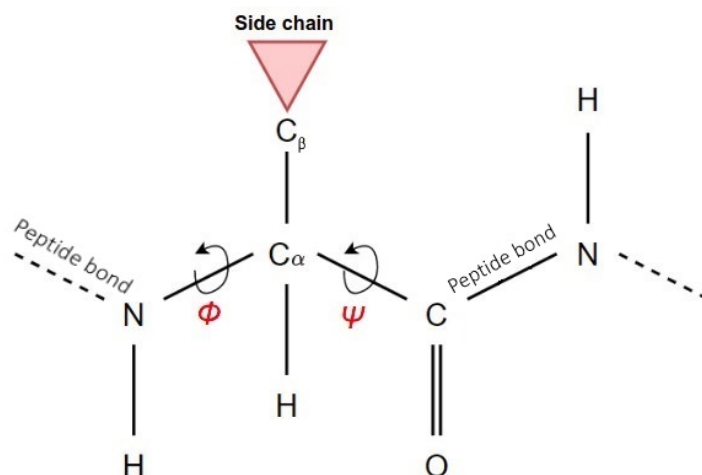


Figure 1.1: Schematic diagram of a single amino acid. It does not apply to glycine which has a hydrogen atom instead of C_{β} and is the only non-optically active amino acid. Dihedral angles are denoted as Φ and Ψ , they are crucial for describing conformational flexibility and geometry of proteins, plotted as Ramachandran plot, dihedral angles reveal type of secondary structure of the backbone.

precisely regulated physical process known as protein folding. This process occurs spontaneously and is divided into main four stages, the first one being primary structure, which is dependent only on linear amino-acid sequence. This specific arrangement of amino acid residues determines the proteins native conformation. The secondary structure begins to emerge as certain local structures, such as alpha helices and beta sheets, form through hydrogen bonding between nearby amino acid residues. The tertiary structure of a protein

is the three-dimensional arrangement of the entire poly-peptide chain. This structure is achieved by an **increase in entropy** as seen of 1.2 and stabilized by various interactions, including: hydrogen bonding, Van der Waals forces and disulfide bridges[3]. Finally, the highest classification level of protein structure (quaternary) is reached when two or more smaller protein chains, called subunits, are joined to form a larger multi-subunit complex. The final 3 dimensional structure determines function of the protein, which can be astonishingly complex. A capsid is the protein shell, which encloses the genetic material of a virus. It consists of repeating subunits joined together, called capsomers. It serves not only protective function within the virus, defending it from chemical and physical threats, they also interact with the host's cells, leading to penetration of cellular membrane and infecting it with viruses genome. The capsid of HIV-1 virus (CA) comprises of 1500 copies of virally encoded protein, arranged in a hexagonal lattice to form the capsid shell[15]. The former one is the most abundant protein within the virus, known as p24. It is made up of 7 α -helices and β -hairpin with a molecular weight of 24 kDa[24].

To better understand protein dynamics, we have to make notice of the Levinthal's paradox. The dihedral angles within an amino acid have 2 degrees of freedom 1.1, therefore an average protein occurring in human body consisting of 480 amino acids [3] the resulting number of possible conformations is astonishingly huge, 3^{478} different states, meaning if the protein was exploring the states randomly, it would take longer than age of the universe to achieve its native form [25]. It is now widely acknowledged that proteins do not exist in a fixed, regular structure. Instead, they form a dynamic collection of conformations spanned across a free energy landscape, influenced by the Boltzmann-weighted probability of occurrence. Despite remarkable success in predicting protein crystal structures, such a single-structure formulation of the protein folding problem provides incomplete information about protein function. It is also often found insufficient for structure-based drug design[23]. As discussed by numerous studies [17], protein structures are dynamically modulated by their interactions with small molecules such as ligands. Dynamical protein conformational changes occur on vastly varied kinetic timescales ranging from femtosecond-scale vibrations to millisecond-scale. Understanding these changes is crucial for accurate function prediction [23]

The landscape of conformational space is rough[6], as seen  1.2. It surrounds numerous conformations with similar energy levels and potential transition paths by reason of the multiple degrees of freedom a protein possesses. The conformational ensemble and associated free energy landscape are very condition-specific, reliant on the protein's sequence as well as its surroundings. The former one has a huge impact and encompasses variables like solvent type, temperature, pH level, and ionic concentration. Moreover the conformations can be potentially changing in response to interactions with cellular binding

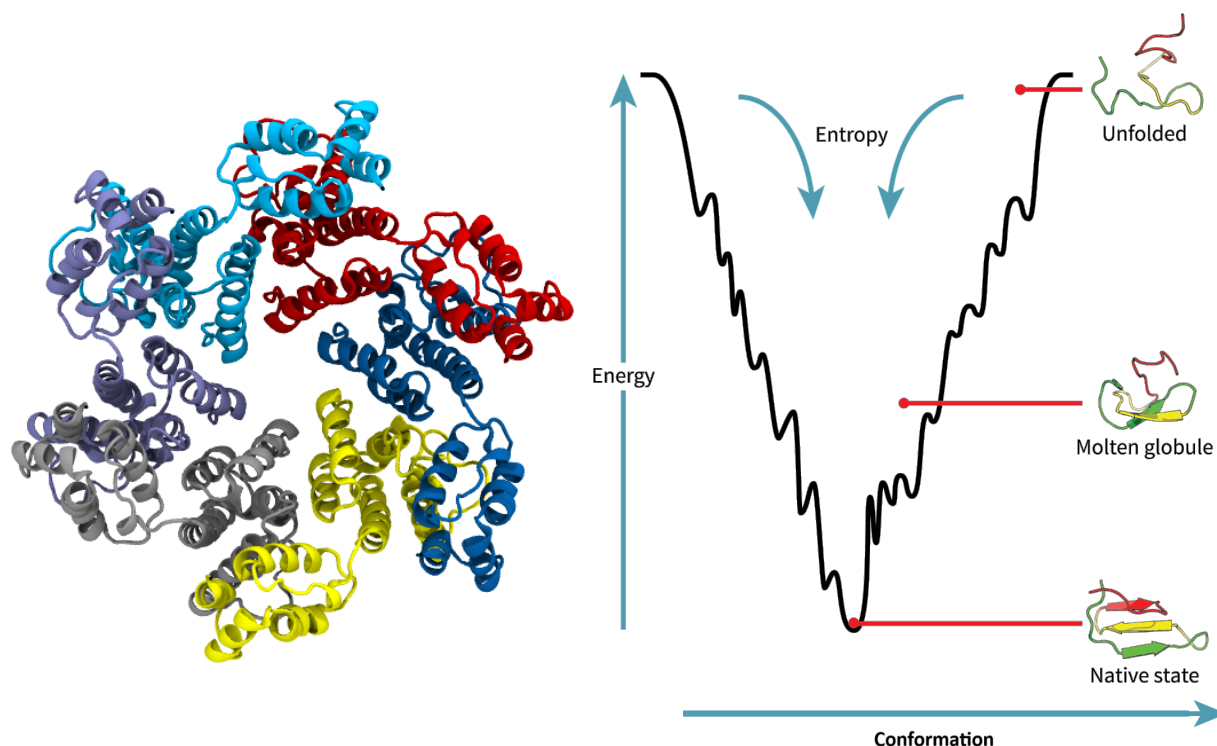


Figure 1.2: Left: Three dimensional representation of *HIV-1* capsomere, where the six monomers forming this complex are shown in different colors. Right: Free energy landscape showing the path a protein takes to arrive at its native state. Higher peaks near local minima represent activation barrier

partners[23]. Understanding the function and ensemble of protein's structures requires a characterization of the free energy landscape and the potential transition pathways, however proteins structure can still be studied using experimental techniques, which require "freezing the molecule" in place, omitting some important data about its structure. Traditionally, the structure is determined using techniques such as X-ray crystallography and NMR spectroscopy, 3D electron microscopy, and more recently X-ray free electron lasers (XFEL)[10]. The most predominant method X-ray crystallography, uses purified and crystallized proteins which are later exposed to X-rays, resulting in characteristic diffraction patterns. These patterns are then analyzed to determine the distribution of electrons in the protein, generating an electron density map used to interpret the location of each atom. This method is very effective however its major limitation is the crystallization process, which poses difficulties while working with flexible proteins, due to the need for molecules to align in the same orientation. On the other hand, NMR spectroscopy is particularly advantageous for studying flexible proteins in solution, offering insights not achievable through crystallography or electron microscopy. In this method, proteins are purified and placed in a strong magnetic field, followed by probing with radio waves. The resulting set of observed resonances is captured and software analyzed to identify closely

positioned atomic nuclei and characterize the local conformation of bonded atoms. These restraints are later employed to reconstruct the full model of molecule. Despite its benefits, NMR spectroscopy is limited to small and medium-sized molecules as a result of overlapping peaks in NMR spectra for larger proteins. While invaluable, these methods encounter challenges in capturing snapshots of protein structures, especially when exploring transient states or undergoing conformational changes. One may inquire about the need for computation when studying protein dynamics, however it proves to be invaluable in studying dynamical systems.

1.2 Molecular Dynamics

The never ending intricacy of nature created beings capable of inquiring about their origin. Our scientific methods have lead us to identify proteins as pioneers in emerging complexity of life, however we are still far away from understanding their nature. To understand the behavior of building blocks of life scientists needed tools that would enable them to look beyond limitations of our vision system, straight into bizarre world of atoms. This revolution was kick started by Alder and Wainwright in 1957 with their paper on phase transitions in hard sphere systems[4]. Their pioneering work, and later contributions led to the development of molecular dynamics (MD), a computational simulation technique used to study behavior of molecules over time. Computer simulations are a very powerful tool allowing us to extensively study systems beyond our reach.

At its core, MD uses Newton's laws of motion and statistical mechanics to predict location of atoms at each time step[4]. Since the fundamental physical laws applied to biological phenomena produce equations that cannot be solved analytically, MD simulations and numerical integration algorithms approximate it as sum of forces associated with bonds, angles, dihedrals, electrostatic and Van der Waals interactions. Ideally, the simulation should last as long as possible to completely sample the conformational space according to ergodic theorem, which states that the time average of states in a dynamical system equals the ensemble average. On the other hand, there is the fundamental concept in signal processing - the Nyquist theorem. It states that to accurately reconstruct a continuous signal, the sampling rate must be at least twice the highest frequency present in the signal. When analyzing protein conformational dynamics, it's crucial to choose an appropriate simulation time step. The Nyquist theorem becomes relevant here, as the time step should be small enough to adequately sample the fastest motions or frequencies in the system[6]. In order to capture biologically relevant phenomena, the timescale of MD simulation must reach order of microseconds, as seen on 1.3, however biologically important phenomena take place on timescales as small as femtoseconds, what poses a

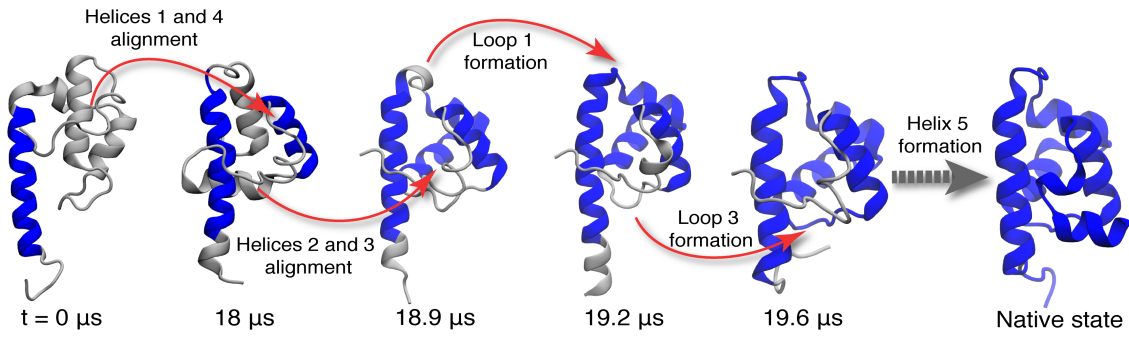


Figure 1.3: MD simulation showing step by step folding process of λ -repressor protein[7]

significant limitation to MD[23]. Furthermore, efficient enzyme catalysis was directly attributed to structural interaction between the enzyme and the substrate via the “lock and key” or the “induced fit” mechanism [10, 17]. However emerging experimental and theoretical approaches provide a revised, broader view in which conformational changes on a wide range of time scales influence or even dominate enzymatic reaction. Improved understanding of enzyme interaction is essential for unraveling the complexity of protein function, understanding disease mechanisms, and informing drug discovery efforts. This is where machine learning offers its promising aspects[23].

1.3 Neural Networks

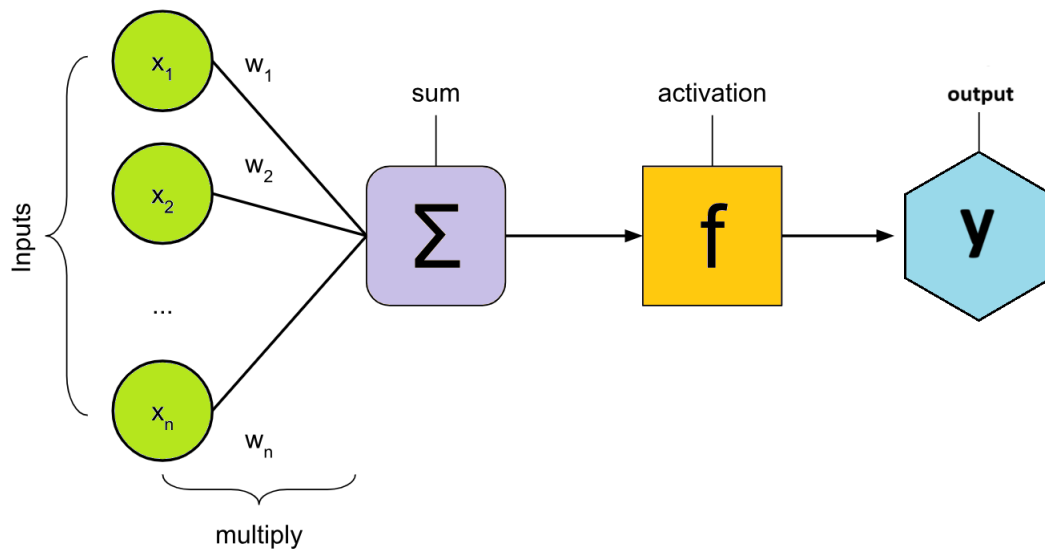


Figure 1.4: Schematic diagram of a perceptron

A mathematical abstraction inspired by interconnected web of neurons in our brain is a powerful tool capable of extracting key features from existing data and using them to

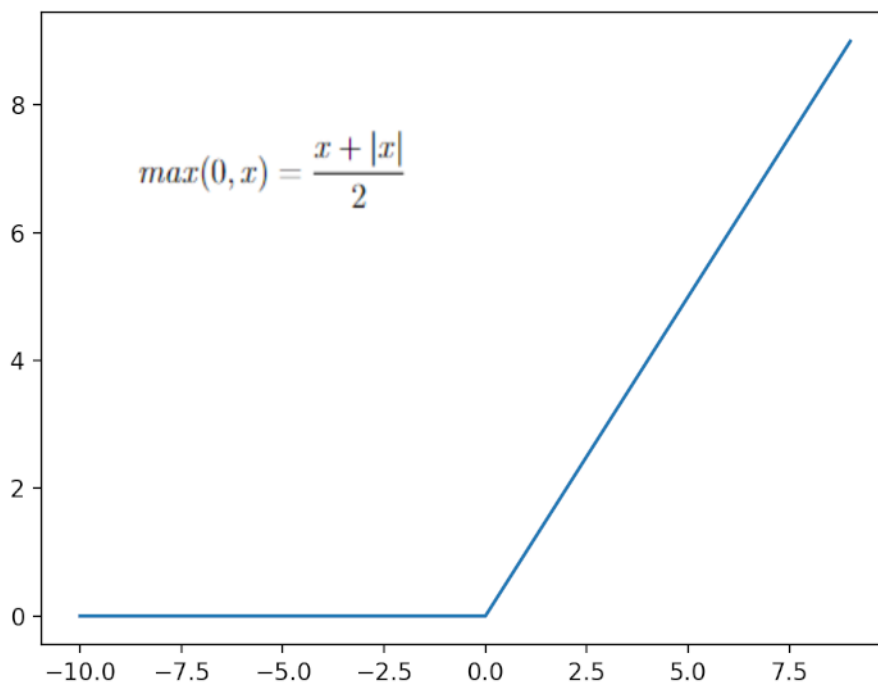


Figure 1.5: ReLU activation function

predict future data or generate new pieces of it. The concept at its core has been theorized as early as 1940s by Warren McCulloch and Walter Pitts, who introduced the idea of artificial neurons[3]. Other milestones in creating today's AI include the perceptron, a type of artificial neuron which gained attention in the 60s through work of Frank Rosenblatt[4], and development of back propagation algorithm in the 80s by Rumelhart, Hinton and Williams. From the beginning of 20th century, advances in computing power and the accumulation of large data sets began revitalization of neural networks(NN), leading to exponential development in the field. Artificial neurons comprise the core of NN, which are layers of interconnected nodes with weights attached, that determine the strength of their connection. The input values and their corresponding weights are summed along with chosen bias and passed through activation function, which is a final way of separating the desired outputs. The most common activation function, also used in this model is ReLU (Rectified Linear Unit), as seen on 1.5. The neural network processes input data through those layers, adjusting the weights, as on 1.4, during training to optimize its ability to produce accurate outputs. Convolutional Neural Networks (CNNs), a subset of deep learning algorithms, are particularly adept at processing grid-like data, making them ideal for applications in scope of this project. To study protein's conformational space, CNNs and their application in autoencoders provide sophisticated set of tools for extracting meaningful representations from complex, high-dimensional data[6]. CNNs have been primarily developed for image recognition and computer vision [11], however their ability

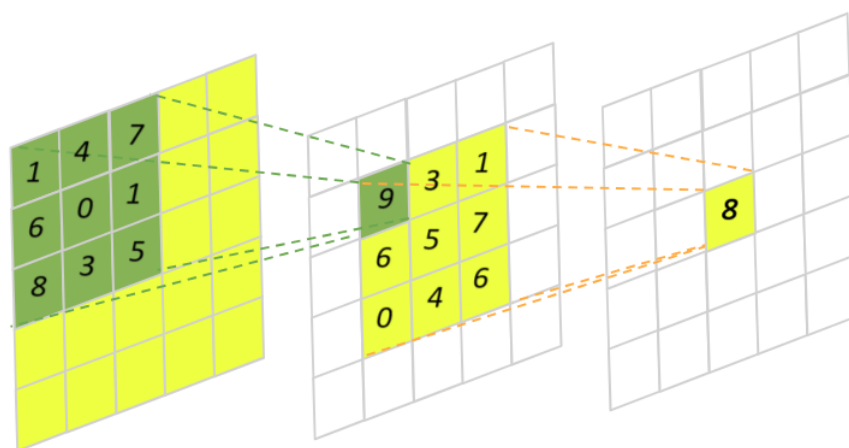


Figure 1.6: Kernel sliding across variable matrix extracting next layer data

to take multidimensional inputs without changing structure of the network makes them well suit for protein dynamics[2].

Convolution is the process of adding each element of the image to its local neighbors, weighted by the kernel 1.6. This is related to a form of mathematical convolution. The matrix operation being performed is not traditional matrix multiplication, rather Hadamard product, a binary operation that takes in two matrices of the same dimensions and returns a matrix of the multiplied corresponding elements. Unlike traditional matrix multiplication it is commutative. This operation leads to reduced dimensionality of data, layer by layer leading to 2×1 matrix Machine learning has been quite impactful in the analysis of protein dynamics[6]. In this context, two closely related aims are reached, firstly the extraction of collective variables (CVs) associated with the slowest dynamical processes and the metastable states from given protein molecular dynamics simulation data [6]. Secondly, NN have capability of enhancing the simulations so as to increase the number of rare event transitions between them.

Input data for the neural network is obtained through MD simulation, and stored in .pdb files, where position of each atom in 3 dimensional Cartesian space is recorded over time. The process is repeated for multiple conformations resulting in vast data space. All the atom coordinates are fed into the neural network. The analysis is performed in each ~~perceptron1.4~~, where the inputs are summed and a suitable bias is added. The resulting value is passed on through the activation function 1.5 which decides whether the perceptron will 'fire' or not. The resulting output is then processed by a loss function, where it is scored against the correct answer from the validation set. The loss function is desired to be minimised in procedure called back propagation, where weights of each connection along with biases are finely tuned to match the expected output.

2 Project Objectives

As shown by Degiacomi research group, **Molearn** has capabilities to generate transition paths and new conformations of rigid proteins[14]. We have all seen what neural networks are capable of, 2.1 .The objective of this project is to test effectiveness of Molearn with larger, multimer proteins and eventually improve its performance. Firstly, the software is tested with **MurD**, to see how large gaps in Cartesian space between adjacent subunits of the capsomere, affect performance of Molearn. MurD is deliberately chosen as the benchmark case for this experiment, as it is well studied by the Degiacomi Group. The hope is that Molearn will be able to accurately reconstruct rearranged MurD molecules, with low RMSD difference between input and decoded data. When this evaluation is complete the network will be trained with p24 protein, a new challenge for Molearn. Performance of the network will be evaluated against different testing criteria, leading to an improved model. In the end the training is repeat with different dimensional sizes of the latent space to see which one performs the best.

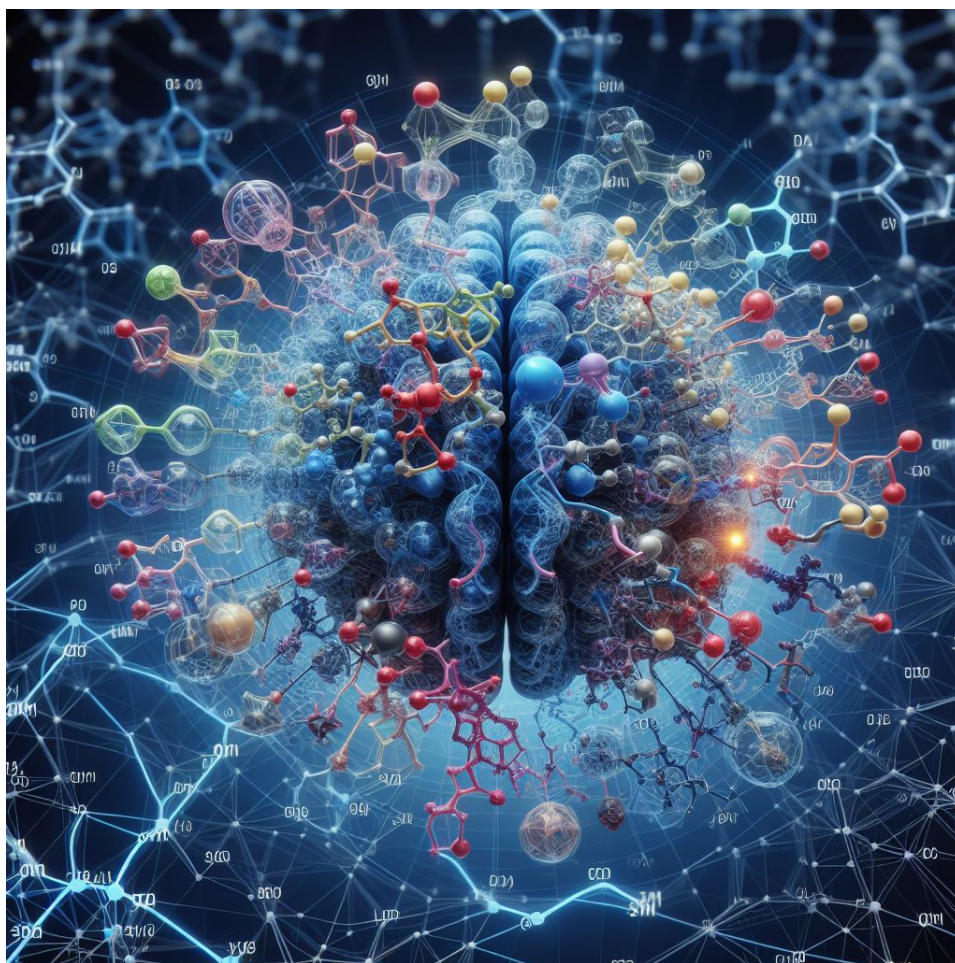


Figure 2.1: image of a neural network learning protein structure generated with DALL-E3

3 Materials and Methods

3.1 Molearn

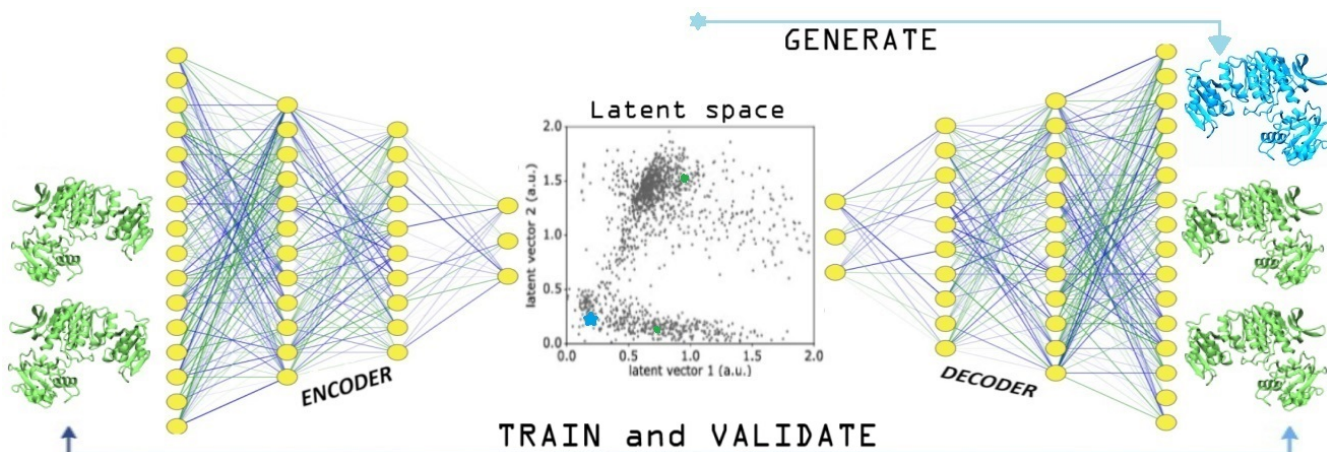


Figure 3.1: The encoding part of an autoencoder schematically presenting how atom coordinates are extracted through ~~convoluted~~ layers leading to a point in 2 dimensional latent space

 Python package developed by Degiacomi research group, based on a deep generative autoencoder, which allows for accurate transition paths prediction in rigid proteins [18]. In terms of network architecture, the encoding and decoding parts of the neural net are symmetric upon mirror reflection, with 2 dimensional latent space located between them. Each of the CNN parts consists of 6 convolutional layers followed by max-pooling layer and a kernel of size four. This neuron setup learns 32 feature detectors, with stride equal 2 it halves the size of input data layer by layer, transforming the protein into more abstract, ribbon-like structure[18]. Before being encoded as a single point in latent space, the protein is reduced to n -dimensional tensor. Using convolution means the same network architecture works with input molecule of any length. First part of this **NN** architecture encodes each protein as a single point in the latent space, as seen on 3.1 Decoder takes this point as it's input and attempts to recreate the original protein. This output is then passed onto a loss function, which aims to minimise input-output variance and defines the learning rate. The loss function in Molearn couples two distinct terms, geometric and physics-based. The former aims to minimize the mean squared error of input and it's decoded counterpart. The latter ensures that latent space interpolations between any pair of conformations produces structures of low energy. The autoencoder takes as input a matrix of backbone atoms including C_β , these are $[N, C, C_\alpha, O, C_\beta]$. The list of respective atoms with their 3 dimensional position and index within the chain are stored as .pdb files.

Protein Data Bank (PDB) is an online repository containing 3D structural data for large proteins. Established in the early 1970s, it provides worldwide access to experimentally determined structures of these biomolecules. Moreover, PDB has its own file format for storing protein structural data, which is used in this project.

The bottle neck of an autoencoder, also known as the latent space is the key to correct functioning of our model. It is here, where if the network is sufficiently trained, low dimensional representation of set of protein's conformations are stored[5]. Creating such landscape allows us to find transition path between different states, study meta-stable conformations or even extrapolate to generate new, unseen ones [18]. It is worth noting, that due to decreased dimensionality, the transition paths take form of straight lines in the latent space, when in higher dimensions they form curves, encompassing many possible transition states., as shown on 3.2

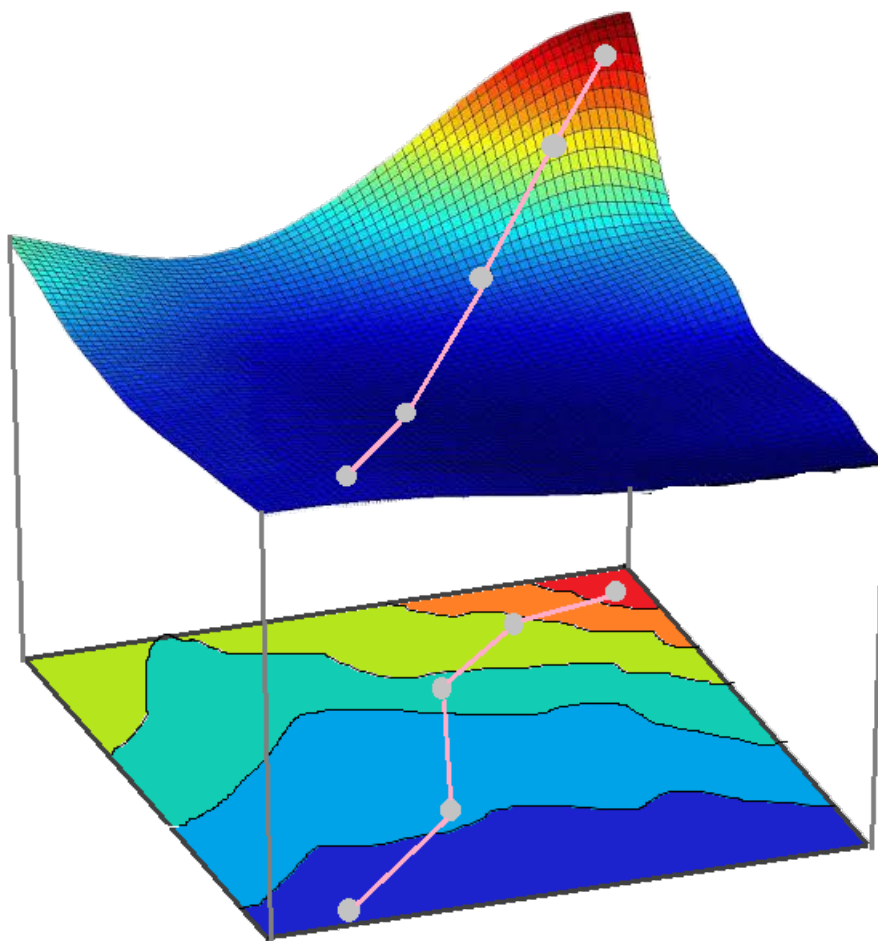


Figure 3.2: Latent space of autoencoder projected onto more dimensional surface presenting it's rigged appearance. The solid line symbolizes example transition path

3.2 Biobox

Biobox is a Python-based toolbox which integrates biomolecular modeling by simplifying the manipulation of molecular structures and electron density data [19]. It has been instrumental in studies of assembly of protein complexes. It is used in my research for efficient protein structure manipulation, like assembly of rearranged MurD molecules and p24 multimers.

3.3 DOPE score

In ~~principal~~, the quality of a generated protein structure can be assessed by sampling the potential energy surface [20]. This approach is computationally prohibitive, instead alternative solution is used. A scoring function whose global minimum corresponds to the native structure of protein from a sample of native structures with different sequences [20]. Discrete Optimized Protein Energy (DOPE) is an atomic, energy based, distance-dependent scoring function, grounded entirely in probability theory [20]. One of its main benefits is identifying physically plausible structures among decoys. The lower the DOPE score of a generated structure the better its quality. Limitations of DOPE score include not integrating solvation effects and ligand interactions, however molecules used in this research are well suited for its application. DOPE score is implemented in MODELLER python library and used throughout this project.

3.4 Computation

To conduct ~~relative~~ research for the project, substantial amount of computation was handful. It was provided to me by the Durham Department of Computer Science. I was granted access to the NCC (NVIDIA CUDA Centre) shared research GPU cluster, what made the required computing possible. The NCC system comprises ~~the~~ head nodes, twelve GPU compute servers, and six CPU blades, totaling 82 GPUs and 246 CPU threads. The protein structure (.pdb) files were of size ~1 Gb and training the network with them would take around 30 minutes per structure. Moreover, even with high performance computing, analysing the network checkpoint files would take between an hour and two.

4 Results and Discussion

4.1 MurD

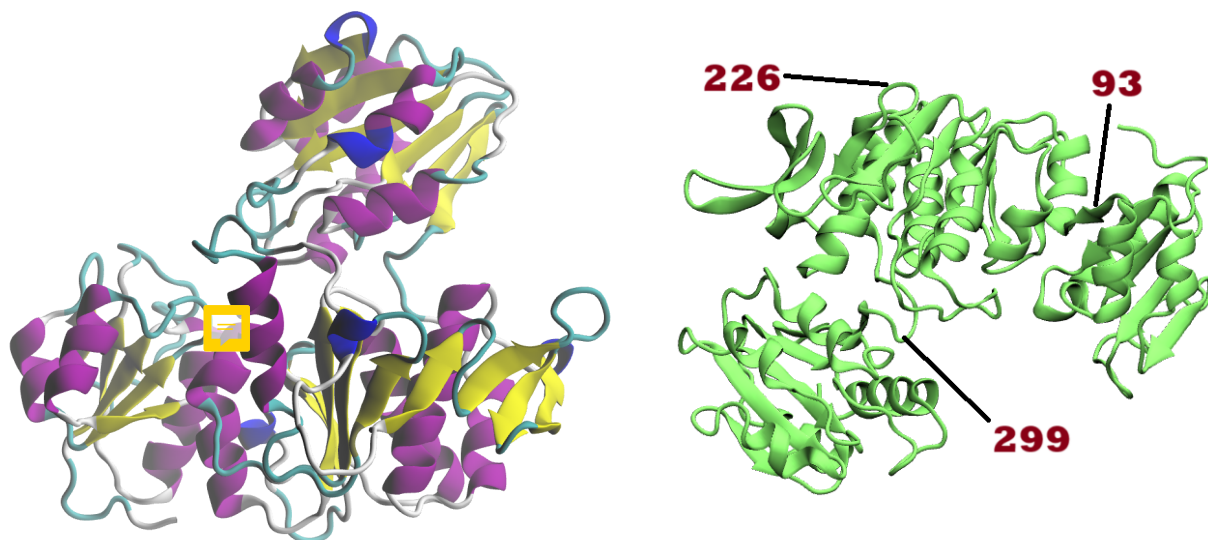


Figure 4.1: Left: secondary structure representation of MurD molecule. Right: Cartoon representation of the same molecule with residues where the protein is cut labelled.

To test whether Molearn can be trained on the large capsid protein it was first trained with a smaller, more researched protein, MurD. Degiacomi research group have studied it with Molearn extensively, so it ~~seem~~ well fit for a test case. Firstly, MurD was modified in order to resemble a multimer a little bit more. Multimers structure, saved in *.pdb* files differs from a monomer, as atom coordinates are in the same order as in the respective subunits, however atoms from the end of subunit 1 and beginning of subunit 2 are in consecutive rows of the *.pdb* file, when in Cartesian space they are considerably separated. This fact poses difficulty to the trainer of the neural net, which relies on physical distance between the atoms to find the right model [9]. The investigation was begun by creating artificial 'MurDmer' from one molecule. It was achieved through rearranging MurD molecules with Biobox to create synthetic gaps between adjacent atoms in the chain. To do this three copies of the molecule, were cut at residues 93, 226 and 299, presented in fig4.2 Halves of the respective proteins are then swapped and rejoined for each of the 4228 conformations. The new, rearranged molecules were used as input for the neural network. First, the files were trained using `OpenMM_physics_trainer` which would result in many errors raised by `MODELLER`. This trainer was used before on other molecules, however it was not able to evaluate force field of shuffled MurD, because of the large coordinate gaps between adjacent atoms. Training each of them took about 30 minutes and checkpoint files were later analysed.

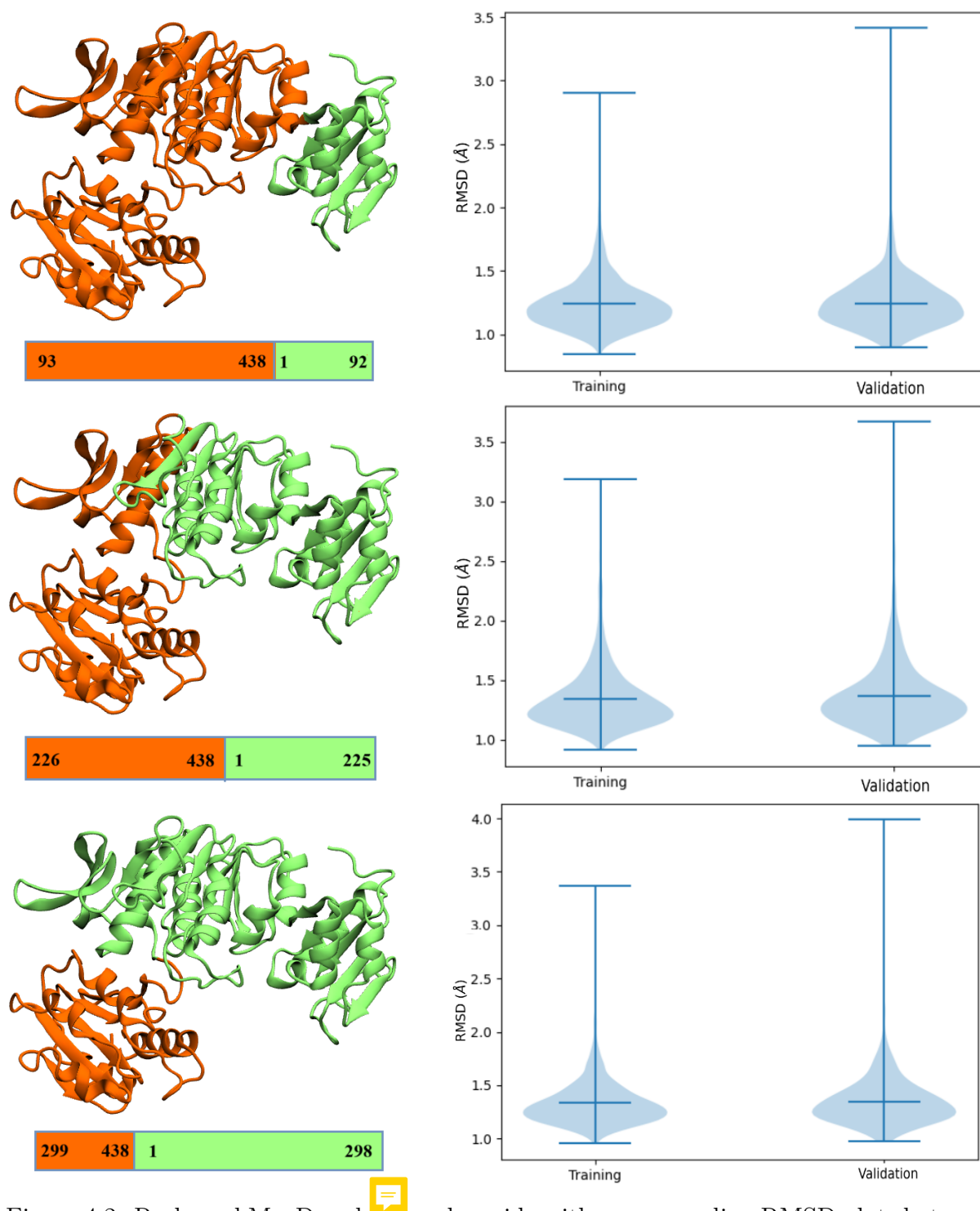


Figure 4.2: Reshaped MurD molecules alongside with corresponding RMSD plots between input and decoded data for training and validation sets. The plots showcase low RMSD for each of the molecules trained, meaning Molearn can handle coordinate interruptions

4.1.1 Latent dimension and reconstruction accuracy

In this section we examine the critical influence of a latent space in the performance of our autoencoder model to accurately reconstruct unchanged as well as rearranged MurD

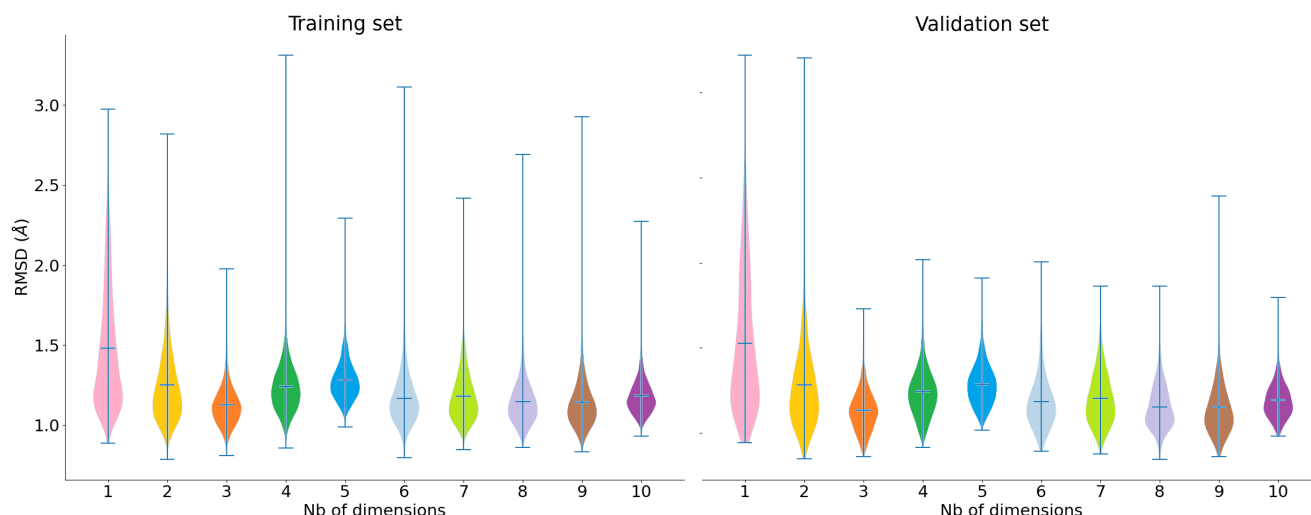


Figure 4.3: MurD structure reconstruction error with increasing dimensions of the latent space.

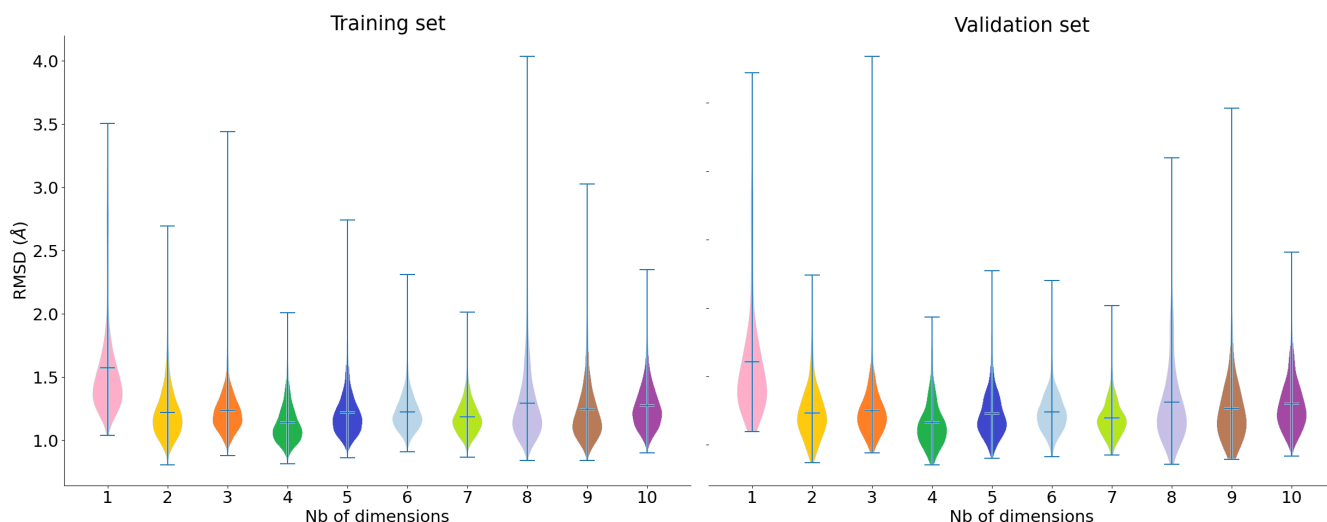


Figure 4.4: Structure reconstruction error with increasing dimensions of the latent space. This time the network was trained on rearranged MurD, cut at residue 226 and the halves were swapped.

molecules. The bottleneck of our model is an influential factor to the accuracy of network output. The model used extensively to study MurD, features a 2 dimensional latent space, giving enough accuracy to study molecules of similar size. Given the complex structure of protein configurations, determining the optimal latent space size is essential to balance the trade-off between sufficient data representation and overfitting. Therefore, to efficiently use Molearn with multimers, other latent dimensions need to be taken into consideration.

The unaltered MurD molecule was used for training the neural net with different latent dimensions, results seen on 4.3. The plot showcases there is no clear trend that suggests increasing dimensionality consistently reduce the reconstruction error. The variability

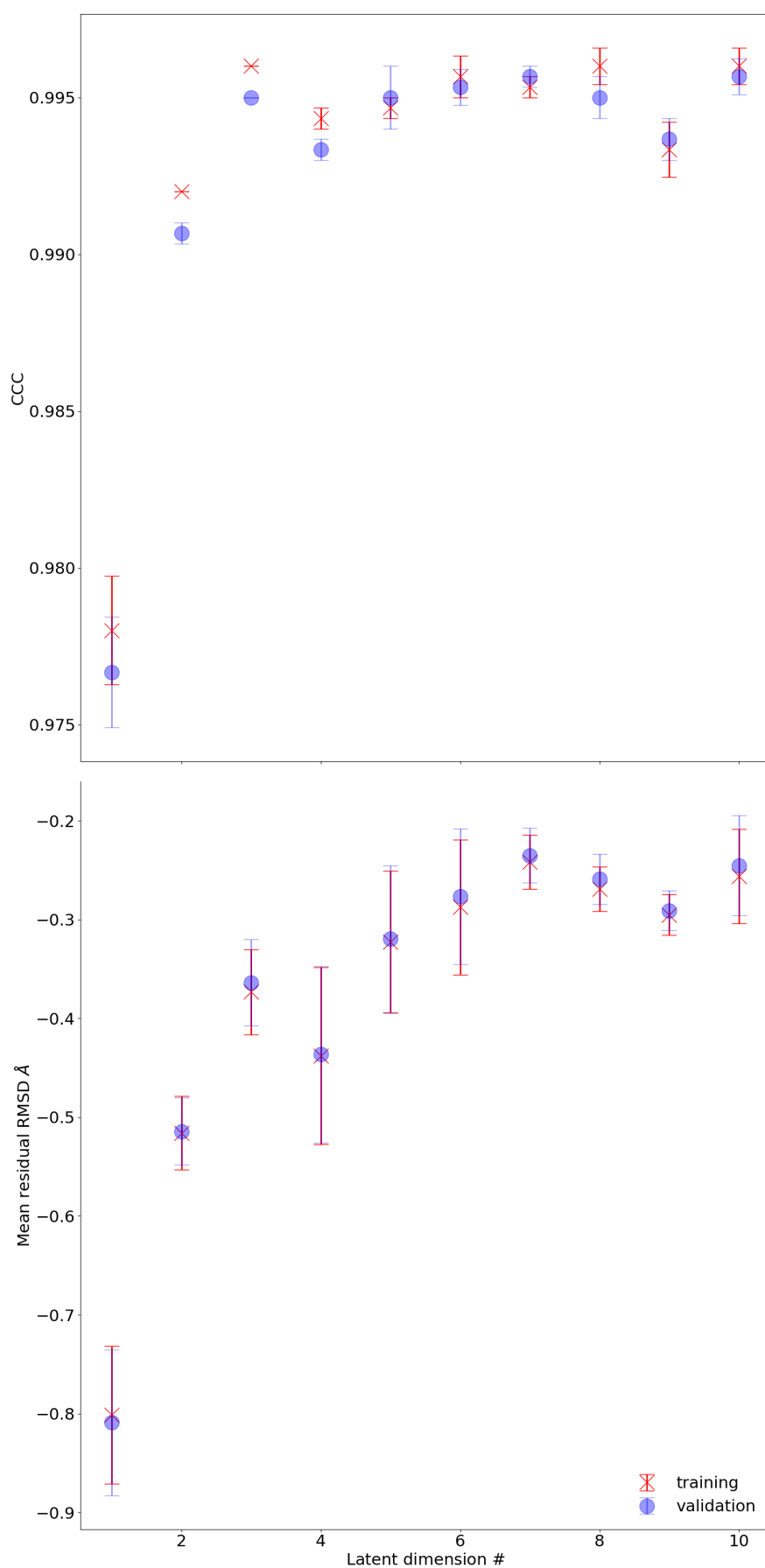


Figure 4.5: Cross correlation coefficient sharing axis with mean residual RMSD against respective latent dimensions computed for the original MurD molecule, relevant at the end of this section. Both scores on the ordinate were averaged over three runs, qualifying for standard error bars. Both metrics show that dimension 7 performs the best with this small molecule

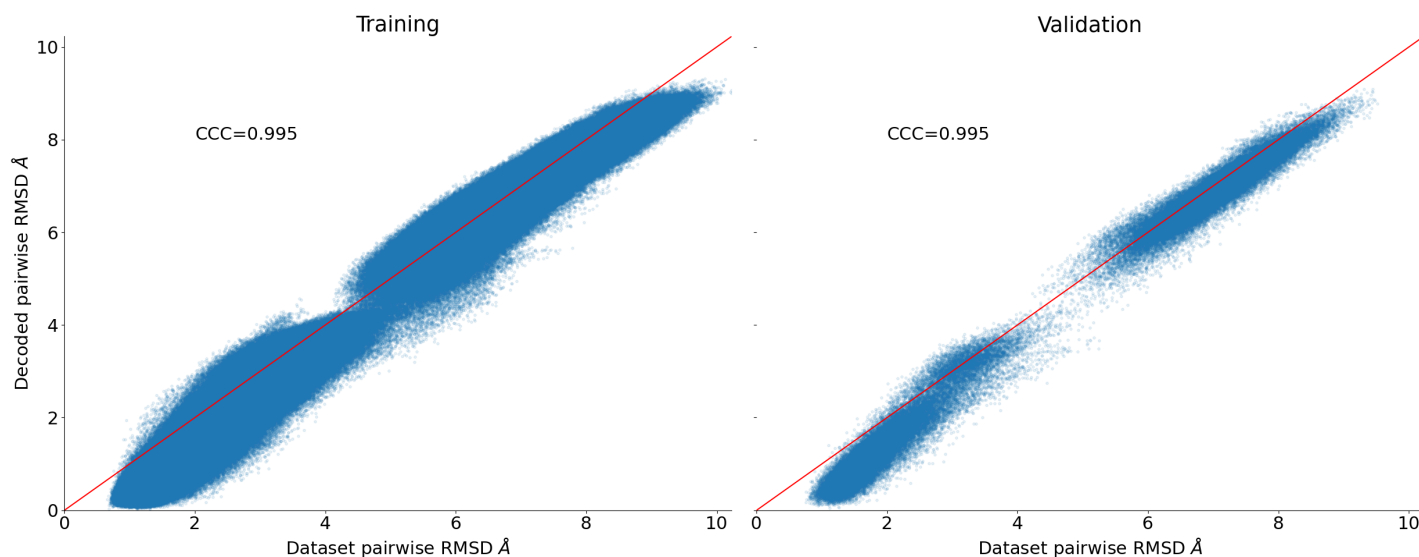


Figure 4.6: Decoded against dataset pairwise RMSD with 10 dimensional latent space. The data is split into two regions of points according to RMSD of bound and open MurD conformations

of RMSD is coherent all through the figure, with some dimensions producing more, less precise structures. To investigate which latent size is the most optimal for training with p24 multimer, the same computation was performed with shuffled MurD structure. The molecule chosen for this was the one cut at residue 226, as it is approximately the mid residue of MurD. The results of this experiment, shown on 4.4, are very to the previous trial 4.3. The reconstruction error fluctuates without a clear pattern as the number of latent dimensions increases, however this time, second dimension produces fewer outliers than when training with original MurD. On the other hand, these plots were redone 3 times, however 4.3 is not an average of those runs. These results set a guideline as to what we can expect when investigating latent dimensionalities further. To do so we used Biobox function `rmsd_distance_matrix` which returns a matrix of pairwise RMSD structure differences within the sample. This test was performed on the dataset ensemble (creating a benchmark for comparison), as well as on the set of decoded structures. 4.5 shows extensive analysis of different latent sizes to find the one best suited for MurD. The plot 4.6 shows a matrix of RMSD differences between structures within each of the two sampled sets. The diagonal red line on the plot symbolizes a perfect scenario, where the network does not overfit the data and the decoder perfectly reconstructs each coordinate of the latent space to a full protein. The same type of plot was obtained for each dimension three times to ensure consistency, however only dimension 10 is shown in report for space preservation. Additionally, cross correlation coefficient was calculated for each of the plots and is labelled as 'CCC'. This index measures how well are the variables correlated,

indicating strength and direction of their linear relation. This kind of analysis has not been performed before by Degiacomi research group. The points on 4.6 very nicely aligned with the diagonal red line, showing that 10 dimensional latent space is performing well. It is worth noticing that larger portion of the data is below the line, meaning the average pairwise RMSD is larger within the dataset than the decoded set. This is due to loss of variety of the original data. In order to produce more precise results, the structures decoded by our model are generalised. This cunning practice of the NN means the diversity of our dataset is lost and replaced with averaged structures[8].

4.2 HIV-1 capsid protein

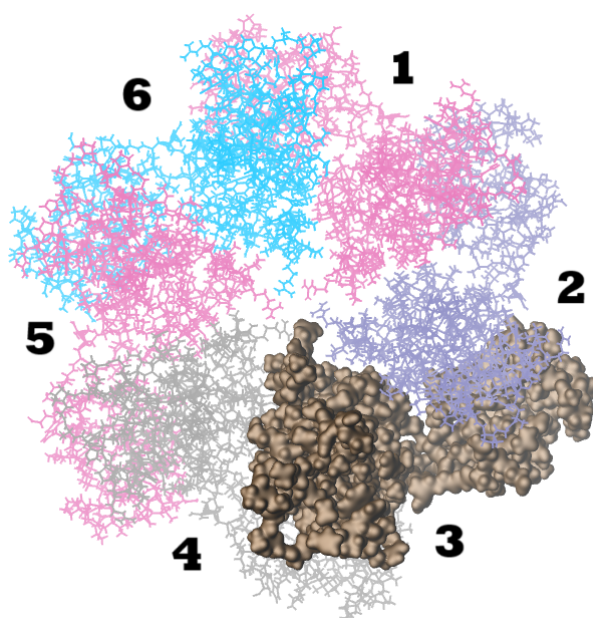


Figure 4.7: The hexameric HIV-1 capsomer, made up of 6 identical monomers. This is the 3MGE crystallized structure of p24 protein. Chosen representation depicts each bond within the molecule alongside an isosurface representation (shown in brown) distinguishing regions of the monomer with the same electron density

After successfully confirming that the Molearn trainer can achieve sufficient accuracy in reconstructing shuffled MurD molecules, the same network was trained with the p24 hexamer 4.7. Firstly only single subunit was introduced to the network, results seen on 4.8. The training was performed only with the standard mean square error loss function (MSE), and the results are precise and accurate. The interquartile ranges (thick part of the violins) are relatively tight, indicating the former. The training and validation are randomly split is 9:1, with constant seed throughout the project, resulting in the validation set performing slightly worse than training consistently, as expected. In terms

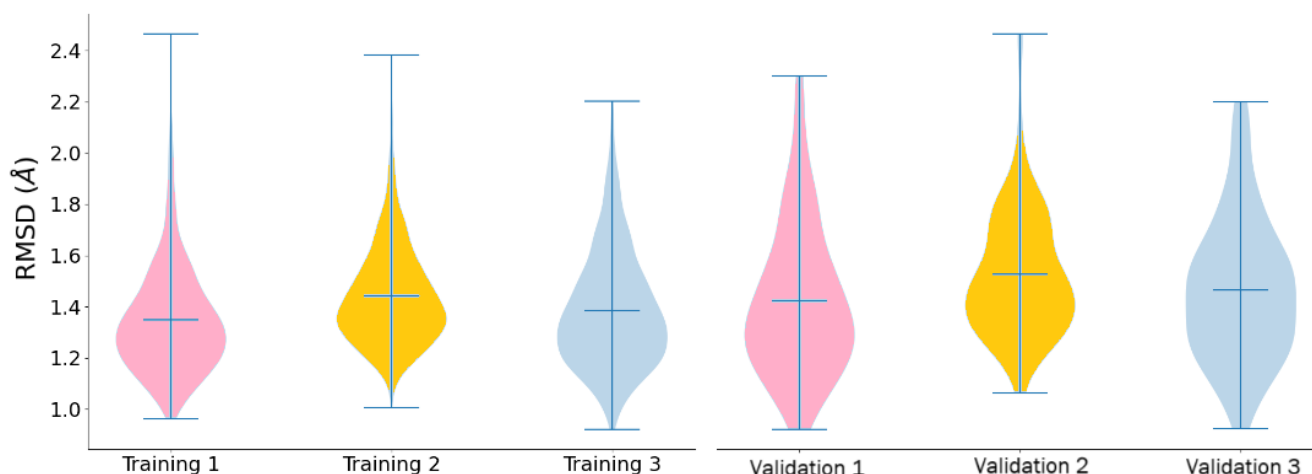


Figure 4.8: Reconstruction error of the p24 monomer dataset decoded and encoded by a trained neural network. The results are consistent for all three trials with average RMSD of 1.4Å

of accuracy, the goal is for RMSD to be as low as possible. With the median score around 1.4 Å 4.8 the results are reasonably accurate. On the other hand, there are outliers (indicated by extended arms of the violin), implying the variability in each run. It is mainly caused by random initialization, meaning the random initial weights chosen by the network, which can lead to different optimization paths and convergence points[5]. This training as well as the others, was redone three times for a few reasons. Firstly to ensure consistency of our results and secondly to avoid local minima. These are points in the model's parameter space where the loss function has a value lower than in the closest surrounding, but it may represent the global minimum[16]. Moreover, when training without the physics loss function the network is prone to overfitting. If the network's accuracy on the training set is significantly higher than on validation, or when RMSD values of validation increase while they decrease for training, the network is overfitting. To prevent that, each dataset was split into training and validation sets. After achieving plausible results with the monomer, other subunits were introduced to the NN and the results are shown on 4.9. The figure clearly shows how RMSD of the structures is directly proportional to number of subunits being training i.e to number of atoms present in the dataset. Range of the RMSD values for each consecutive number of subunits shows variability in the data. It is worth noting that on 4.9 the violin's range is uncorrelated with the number of atoms present, meaning variable performance and lack of correlation with size of dataset. On the other hand, both training and validation sets are fit with a trend line, shown in red. The close similarity of both lines suggests consistency of the model across both sets. Gradient of the validation slope (0.447) is within 3% difference from the training set, while the difference in $y - intercepts$ of both lines is within 10%. The constant part of the equations can suggest the baseline level of error for this

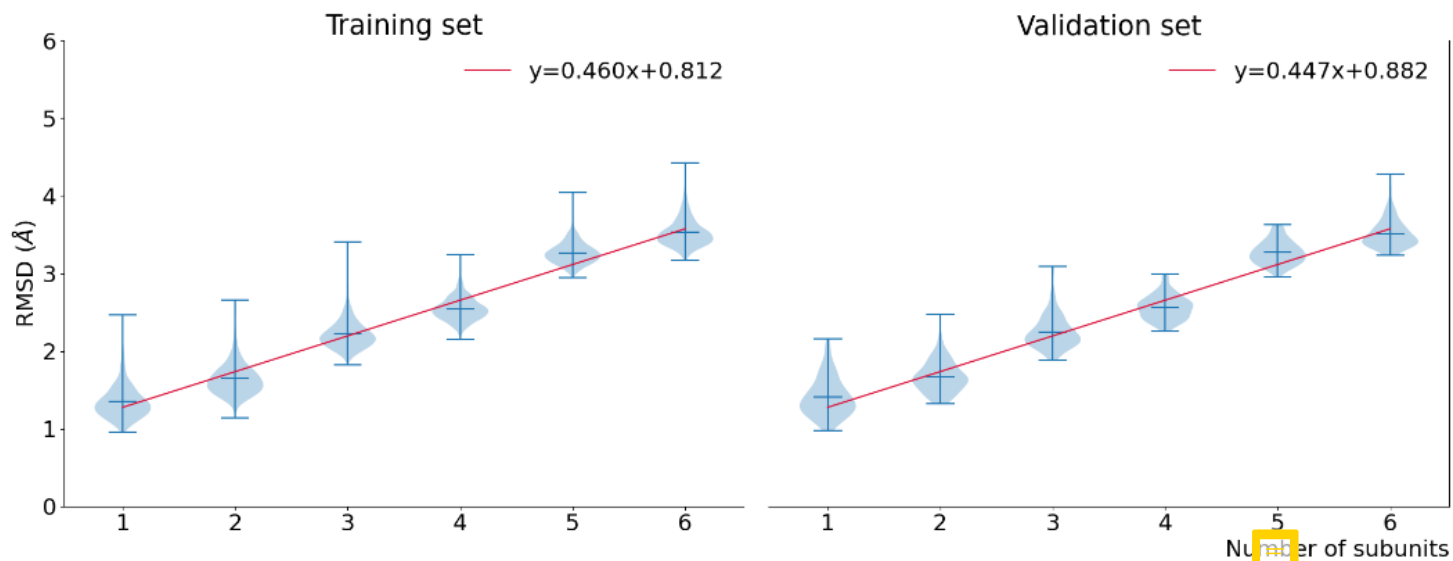


Figure 4.9: Training and validation sets of p24 protein with increasing number of subunits

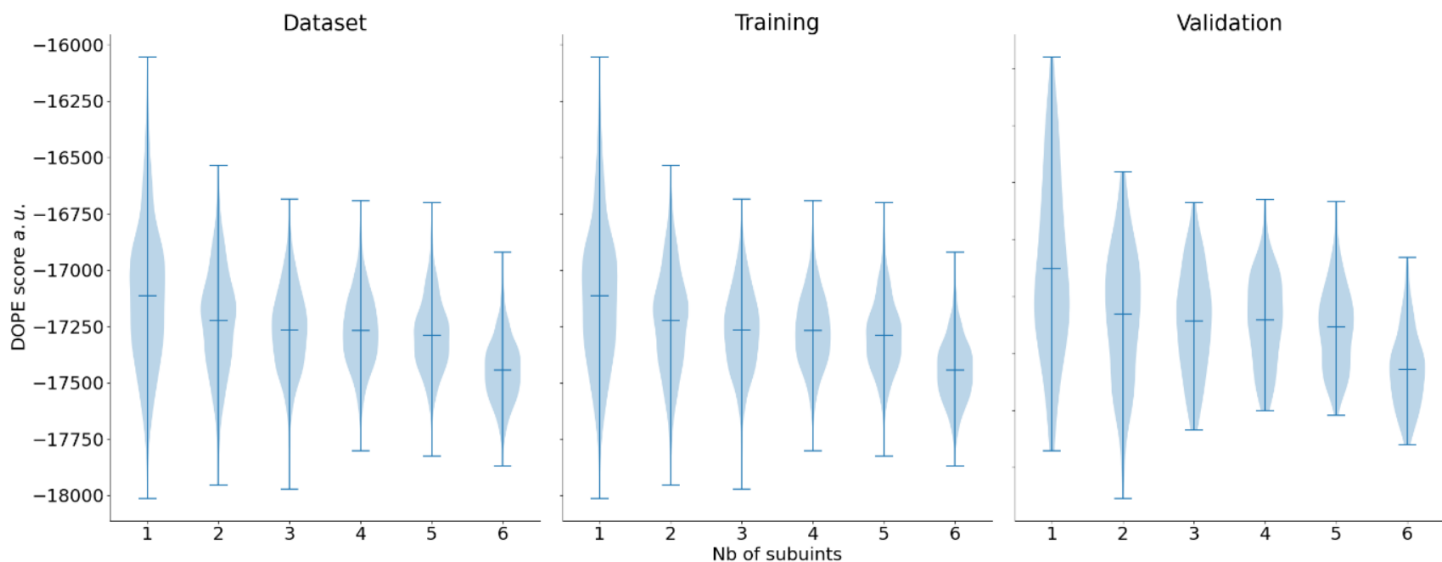


Figure 4.10: DOPE score per subunit of the p24 hexamer.

network, when comparing small proteins. As number of atoms in the sample increases (by training with additional subunits) and the latent dimension is constant, the data compression factor increases, resulting in less accurate structure prediction by the neural network[13]. When latent size is increased, the compression factor decreases, indicating that more dimensional bottleneck of the autoencoder can produce more accurate output. This hypothesis is soon tested in the rep . To further validate the model, DOPE score was calculated for the dataset, as well as decoded training and validation sets, shown on fig 4.10. The score was computed per respective number of subunits to improve readability of the plot allowing for clearer comparison of physical structure plausibility across the different sample sizes. The dataset score act as a benchmark for assessing the quality of

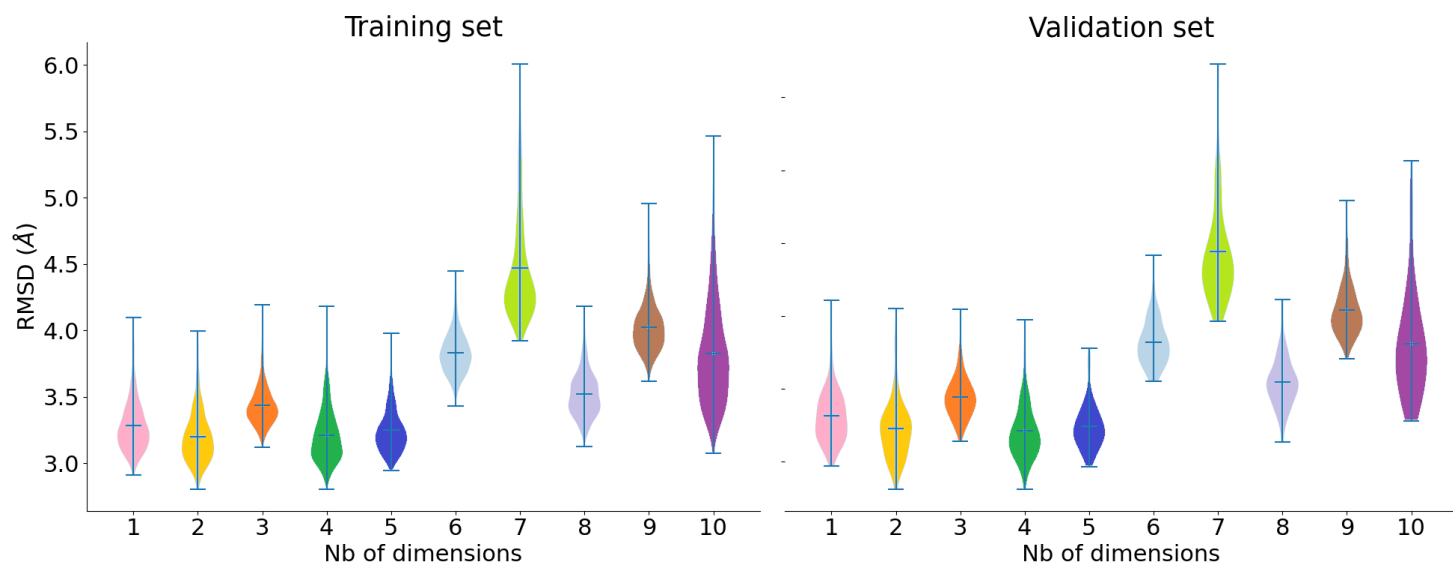


Figure 4.11: Structure reconstruction error with increasing dimensions of the latent space. Training was performed with the p24 hexamer.

decoded structures. The striking similarity between the Dataset and Training sets implies that the network was well trained and could accurately capture the dynamic features of the dataset. Moreover, the last violin on 4.10 is relatively lower than others, for each set. This result is explained by the closed structure of the hexamer, **since it forms a full ring, there is no floppy ends meaning each subunit is equally constrained by its neighbours**. The same principle can be applied to the single subunit, explaining why the violin representing it shows higher DOPE score than others. Firstly, this isolated monomer does not have the stabilizing interactions provided by adjacent subunits, leading to less constraints and more erratic movement. Secondly, values on the 4.10 were divided by number of subunits in the sample, hence only the monomer's DOPE was left unchanged.

4.2.1 Influence of the latent space

As previously done with MurD, latent space of our neural model was reshaped to find which dimension performs the best. Firstly, RMSD reconstruction error was calculated using the `MolearnAnalysis` class. The calculations were repeated multiple times, however results on 4.11 are not averaged. **In** contrast to the MurD test, 4.3, the network trained on the multimer shows narrower spread of RMSD values, as seen on 4.11. On the other hand, the reconstruction error is significantly larger this time, with median values equally spread between 3.2 and 4.5 Å. However, the same values of RMSD were observed on 4.9 when the whole hexamer was trained, meaning the high RMSD values can be attributed to the number of atoms in the system. It is worth noticing how poorly the seventh dimension performed, in fact it was the worst performing dimension in each of three runs, while it was

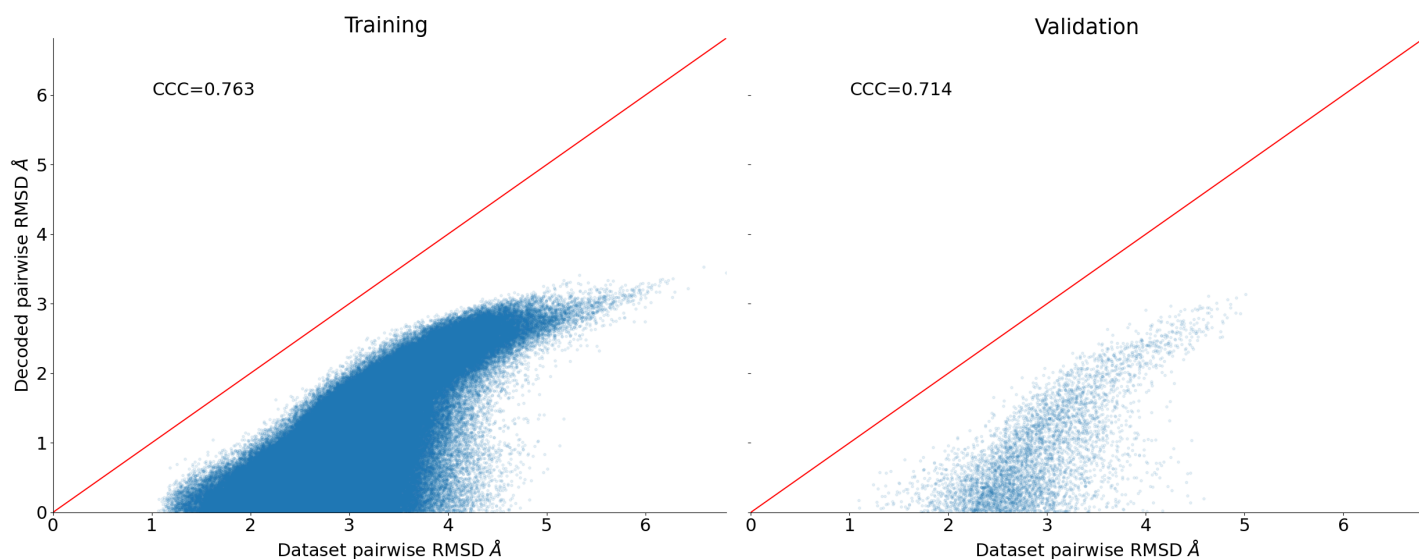


Figure 4.12: Decoded against dataset pairwise RMSD with 1 dimensional latent space. The network was trained with p24 hexamer.

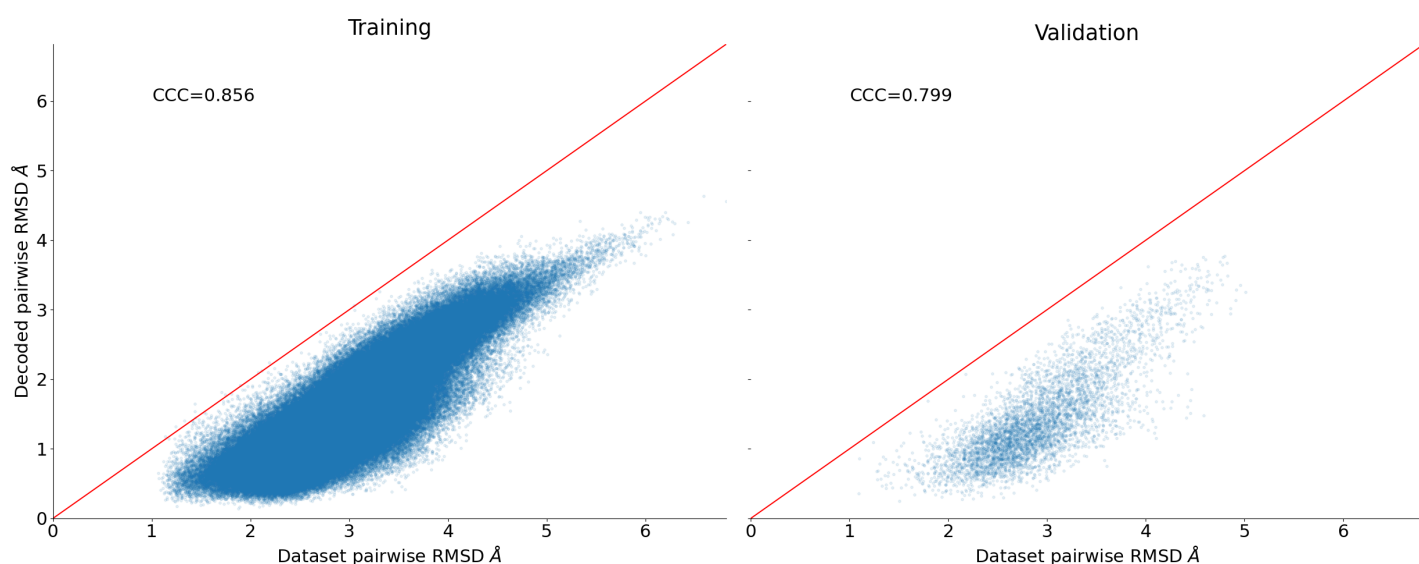


Figure 4.13: Decoded against dataset pairwise RMSD with 10 dimensional latent space. The network was trained with p24 hexamer.

the best performing for MurD 4.5. The decoded against dataset pairwise RMSD test was performed to further validate which dimensions are best performing with p24. 4.12 is the first plot of this kind performed with p24, featuring over 5 million points, computing each of these has taken the NCC 2 hours. All the data on this plot falls beneath the red diagonal line indicating even **greater generalisation** than when training with MurD. This conclusion is further reinforced by the large portion of the decoded pairwise RMSD values, which are zero, or close to it, meaning than many different input structures are decoded into

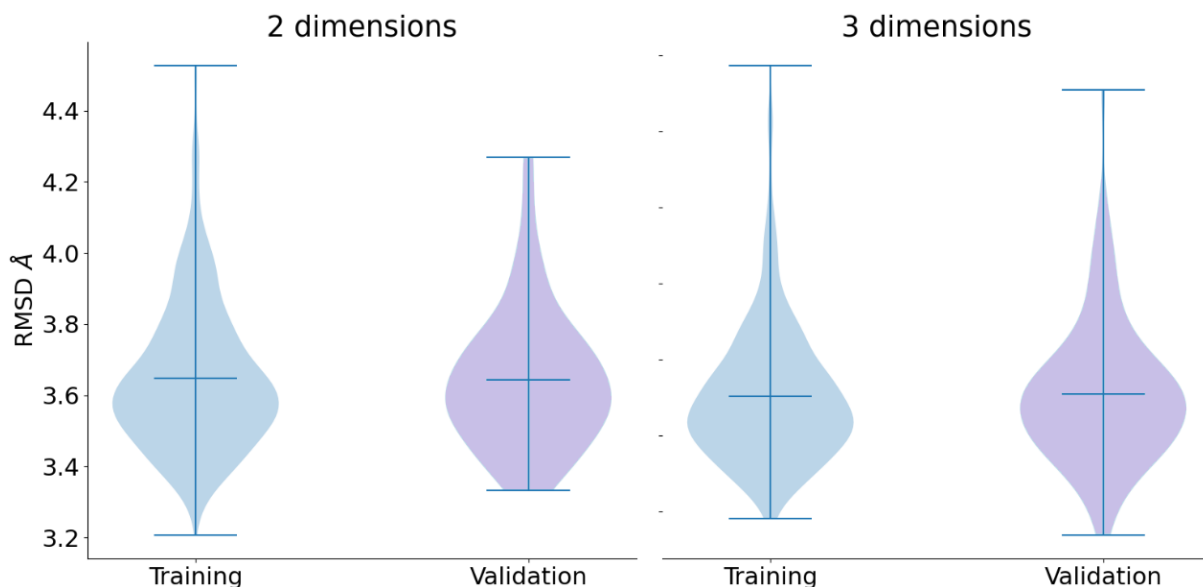


Figure 4.14: RMSD reconstruction error of the p24 hexamer trained with 2 different latent space dimensions

same output structure. On the other hand, the validation set follows the training closely, forming a complementary shape. The same correlation is noticed on 4.13 indicating that, even though the network is generalising the input structures, it is not overfitting to the same degree. 4.13 shows much better diversity preservation of the dataset, as none of the blue points touch the x axis. The cloud of points is still well below the red line, indicating that 10 dimensional latent space is not the best performing bottleneck. To quantitatively find the most efficient latent space each of the decoded against dataset RMSD plots was repeat three times with a uniquely trained network. Next average residual RMSD was calculated for each of the training sets and CCC for every pairwise RMSD figure was recorded, then averaged. 4.15 depicts results of this experiment. Upper part of the figure shows that cross correlation coefficient steeply increases through latent dimensions 1 to 3 showing big improvement in variety preservation. Dimensions 4 and 5 cause sudden decrease of the index most likely due to overfitting. Then the latent size of 6 surprises as the best performing dimension within the set. Very similar pattern is observed for mean residual RMSD on 4.15, except for dimension 4. This latent dimension produced structures with very low mean RMSD, however it remains the only one where validation set performed significantly better to training. This illustrates that latent size 4 may have learned the training data a bit too well, including its noise and outliers resulting in poor generalization to new, unseen data. The best performing latent dimension on the lower figure is again 6, while 4.14 shows slight improvements in reconstruction error when training with the second best performing, three dimensional latent space. Additionally to better RMSD it also provides better overall distribution of structures.

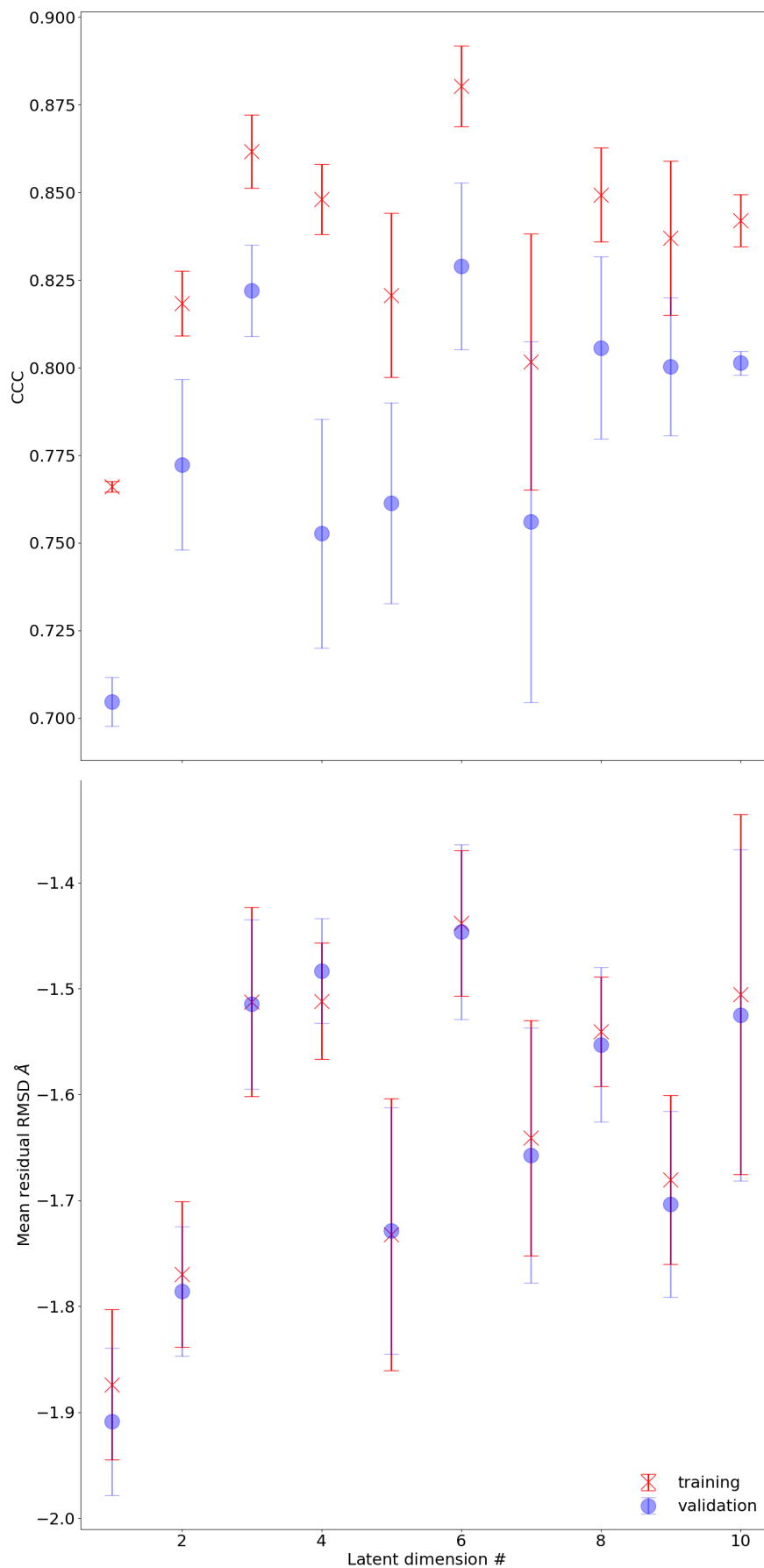


Figure 4.15: Cross correlation coefficient sharing axis with mean residual RMSD against respective latent dimensions computed for the p24 hexamer. Both scores on the ordinate were averaged over three runs, qualifying for standard error bars. Both metrics show that dimensions 3 and 6 perform the best with the multimer

5 Conclusions and future work

In this thesis we explored the function of convolutional neural networks for the complex task of sampling conformational space of proteins. Our journey began with the analysis of well studied MurD and quickly progressed to challenging Molearn’s convoluted training architecture with MurD molecules disguised as multimers. The success in initial experiments laid foundation for tackling more complex proteins, such as the p24 capsid protein of the HIV-1 virus. Our results indicate that despite large interruptions in Cartesian coordinates of adjacent subunits, Molearn can learn the structure and provide a precise output, with reconstruction error scaling linearly with the compression factor. The current network model is capable of reconstructing p24 hexamer conformations with physically plausible  PE scores of -17400 a.u 4.10. Our results suggest that despite achieving valid accuracy, the autoencoder has tendency to generalise diversity of the dataset, averaging less stable conformations into the same output structure. We investigated the influence of latent dimension to balance the precision of reconstruction against the risk of generalization and overfitting. Through this investigation, we have shown that choosing the right dimension of the latent space can preserve more structure variety and decrease overfitting, what is crucial for studying meta stable and transition conformations. Firstly, the influence of latent size was tested on MurD, leading to strong evidence suggesting it the best performing was seven dimensional latent space^{4.5}. Furthermore, this was redone for training large multimers like p24, the best dimensions were found to be 3 and 6, resulting in slightly improved reconstruction error 4.14 and better overall distribution of data, as seen on 4.13

Despite the good reconstruction results, there is still room for improvement in Molearn. Firstly, new promising network architectures keep appearing in the field, giving promising aspects of further development. Recently, the Degiacomi Group has been investigating SIREN architecture, which shows similar results to the current model (~~Foldingnet~~), outputting structures with acceptable RMSD ranges [21]. As more research it carried in the field of machine learning, there are many more promising architectures, like implicit NN [22], providing a novel way to represent 3D shapes. On the other hand, overall network performance can still be improved with the current architecture, by altering parameters like batch size and exploring alternative physics engines (which may be able to handle coordinate interruptions). Fine tuning the batch size may result in faster model convergence, making the computation more efficient, as well as it can influence overfitting and generalization. Larger batch size can stabilize the network against noise in the data, improving the the latter, while providing a more accurate estimate on the gradient, leading to improvement of the former. Constantly pushing the research forward is a difficult task,

however the insights gained from advancements in this field will shape the future of drug design and biotechnology [12].

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5.2 Gen AI statement

Generative Artificial Intelligence was utilized in this project for efficient code debugging and image generation 2.1.

5.3 Summary for general audience

Imagine proteins as tiny, intricate machines within every cell of our bodies, performing countless vital tasks - from fighting off invaders to moving the nutrients around. Scientists have long studied these microscopic machines, but proteins make it more difficult, by being very dynamic. Capturing this array of shapes is a difficult task, however here comes artificial intelligence (AI) and the specialized tools it offers. In this project AI was used to understand how proteins move and change shape over time. It started simple, with a well-research protein, MurD and then progressed into more complex task, p24 protein from the HIV-1 virus. We showed that a Python repository, Molearn, can utilize deep learning on multimers. The research was finished with studying influence of size of the space where data is encoded on the networks overall performance.

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